

# Micromouse Documentation

McMaster Micromouse

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# Chapter 1

## Introduction

### 1.1 What is Micromouse?

A Micromouse is a small autonomous robot that navigates and solves a maze as quickly as possible. The robot begins with no knowledge of the maze layout and must explore it using external sensors. As it explores, it constructs an internal map of the maze, determines the most efficient route, and then completes subsequent runs at increasingly high speeds.

Micromouse has existed as an international engineering competition hosted by IEEE for decades, and remains a great project for learning embedded systems. A successful Micromouse requires the integration of mechanical design, electronics, and software development.

### 1.2 Club Objectives

The goal of our club is to make the process of designing, building, and programming a Micromouse accessible to students of all experience levels. Robotics projects often require knowledge from multiple disciplines, which can make them difficult to approach independently. Through structured workshops, hands-on resources, and comprehensive documentation, the club aims to lower this barrier and provide a clear path from beginner to autonomous robot.

Participants will be guided through the complete development process, from assembling hardware and programming embedded systems to implementing sensing, localization, and path-planning algorithms. Rather than simply providing a finished design, the club emphasizes understanding the engineering principles behind each subsystem so members can modify, improve, and extend their own robots.

While the club provides a complete reference robot, members are encouraged to explore their own ideas and improvements. Topics such as custom PCB design, alternative hardware, and advanced control strategies are beyond the scope of the workshops but are left open for exploration. The provided platform should be viewed as a foundation upon which participants can experiment, iterate, and develop their own designs.

## 1.3 How to Use This Guide

This document serves as both a workshop and technical reference for the Micromouse Club. It is intended to complement our workshops by providing detailed explanation of concepts, design decisions, and implementation details that we may not have time to cover during workshops. It also allows members to revisit or look ahead.

The guide is organized into four main sections:

1. **Hardware:** Introduces the electrical components used in the robot. Explains how they operate and covers wiring.
2. **Software:** Describes sensor integration, motion control, localization, and path- finding algorithms.
3. **Workshop Guide:** Contains material used throughout the club workshops, allowing members to review previous sections or prepare for future ones.

## Chapter 2

# System Overview

2.1 Robot Architecture

2.2 Hardware Overview

2.3 Software Overview

# Part I

## Hardware

## Chapter 3

# Bill of Materials

### 3.1 Components

### 3.2 Selection Rationale



## Chapter 4

# Hardware Theory

4.1 ESP32

4.2 Motors

4.3 Motor Driver

4.4 Encoders

4.5 Time-of-Flight Sensors

4.6 Power System

## Chapter 5

# Electrical Design

### 5.1 Wiring Schematic

### 5.2 Pin Assignments

# Part II

# Software

## Chapter 6

# Software Architecture

### 6.1 Project Structure

### 6.2 Control Loop

## Chapter 7

# Firmware Setup

7.1 Development Environment

7.2 Libraries

7.3 Uploading Code

## Chapter 8

# Sensors

8.1 Reading ToF Sensors

8.2 Reading Encoders

8.3 Sensor Calibration

## Chapter 9

# Motion Control

### 9.1 Differential Drive

### 9.2 PID Control

## Chapter 10

# Mapping and Localization

10.1 Maze Representation

10.2 Localization

10.3 Wall Detection



## Chapter 11

# Path Planning

### 11.1 Flood Fill

### 11.2 Shortest Path Optimization

### 11.3 Alternative Algorithms

**Part III**

**Assembly Guide**

## Chapter 12

# Mechanical Assembly

## Chapter 13

# Electronics Assembly

## Chapter 14

# Firmware Configuration

## Chapter 15

# First Power-On

## Chapter 16

# Testing

## Appendix A

# Pinout Tables



# Appendix B

## Datasheets

# Appendix C

## Equations

## Appendix D

# Troubleshooting

## Appendix E

## Glossary